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VI SEMESTER **B.TECH. [IT]**

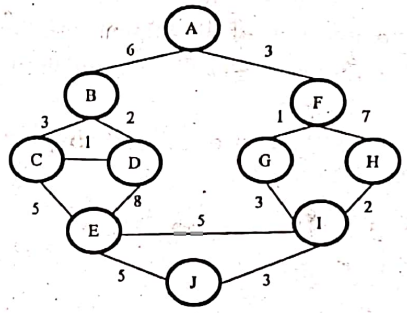
END SEMESTER EXAMINATION **(May- 2019)**

IT-313 ARTIFICIAL INTELLIGENCE

Time: 3 Hours **Max. Marks: 70**

Note: Attempt any 5 Questions.
Each Question Carry Equal Marks.

Q.1) (a) Define the term Heuristic. Find the shortest path from vertex "A" to vertex "J" using A* algorithm. (7)



Node	Heuristic Value
A	10
B	8
C	5
D	7
E	3
F	6
G	5
H	3
I	1
J	0

(b) We have two water jugs; one measures 4 Gallons (4G) while the other measure 9 Gallons (9G). But there is no measuring label mentioned on either of these two jugs i.e. we can't know the exact amount filled in the jug. Now, assuming there is an infinite amount of water supply, explain how we can measure 3G, 5G, 7G, and 8G using

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these unmarked jugs by representing it as a state space problem. (We can empty the jugs, and transfer water from one jug to other). (7)

Q.2) (a) What are agents in Artificial Intelligence? Explain the structure of intelligent agents. (7)

(b) Explain the three variants of hill climbing algorithm: simple hill climbing, Steepest-Ascent Hill Climbing and Stochastic hill climbing. Draw state space diagram for hill climbing algorithm with detailed discussion on different regions in the state space diagram. How the problem of local maximum, plateau and ridge can be solved to find out the optimal solution? (7)

Q.3) (a) Write down the steps of AO* algorithm. Draw AND-OR graph to solve the multiplication problem of 4 matrices and also find out the cost of solving the problem by taking the dimensions of matrices. (7)

(b) Turing test and Chinese room argument are two basic experiments for understanding the concept of Intelligent machines. Explain in detail the practical set-up, process and findings of both the experiments. (7)

Q.4) (a) Find the values of all the alphabets if each alphabet represents a single digit from 0-9 (cryptarithmetic problem) (7)

$$\begin{array}{r} A\ B\ C\ D \\ +\ E\ B\ C\ B \\ \hline A\ F\ G\ A\ G \end{array}$$

(b) P and Q are two propositions. Which of the following logical expressions are equivalent? Justify your answer. (7)

I. $P \vee \sim Q$

II. $\sim(\sim P \wedge Q)$

III. $(P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge \sim Q)$

IV. $(P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge Q)$

Q.5) (a)How genetic algorithm is used for problem solving? Explain with the help of basic selection operators and assuming a suitable fitness function. (7)

(b) How deep learning and machine learning are related to Artificial Intelligence? (7)

Q.6) (a) Write down the steps of Depth First Search Algorithm to traverse a graph, compute its space and time complexity and classify the edges after the traversal as tree edge, back edge, forward edge and cross edge. (7)

(b)Write short notes:

(I) Multi Agent task assignment

(II)Clustering Algorithms

(III)Fuzzy Logic (7)

Q.7) (a) Term Frequency (TF) and Term Frequency-Inverted Document Frequency (TF-IDF) are the two methods in NLP to figure out the similarity between documents. Derive the expression to calculate TF and TF-IDF along with salient features of each one of them. (7)

(b) Explain different applications of Artificial Intelligence in the field of speech processing, Natural language processing and game playing. (7)

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